The Islamia University of Bahawalpur
Department of Information Technology

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E-Commerce Application Development

Chapter
3

Auctions, Portals and Communities

Part I — E-Commerce and Its Models

Taught in Laboratory	Week 2
Assessment	Three shapes compared; your requirements and your decision
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This is a class extract. It is *Chapter 3* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

Dr. Mustafa Hameed · Department of Information Technology

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Chapter 3

Auctions, Portals and Communities

Part 1 | Part 2 | Part 3 | Part 4 | Part 5 | Part 6

Part I — E-Commerce and Its Models

Learning Outcomes

By the end of this chapter you will be able to:

- **Describe** the four common auction formats and **explain** what each one does to a bidder's incentive to bid honestly.
- **Identify** the engineering problems an auction creates that a fixed-price shop does not have.
- **Explain** the role of escrow, and **describe** what a marketplace operator must do when two users disagree.
- **Distinguish** a portal, a marketplace and a community, and **state** where each one's revenue comes from.
- **Explain** the two-sided launch problem and **evaluate** strategies for escaping it.

Before You Start

Chapter 2: the revenue models, especially commission, and the note on network effects at the end.

Key Terms in This Chapter

auction • English auction • Dutch auction • sealed-bid auction • Vickrey auction • reserve price • proxy bidding • sniping • shill bidding • escrow • dispute resolution • reputation system • portal • marketplace • two-sided market • chicken-and-egg problem

3.1 Why This Chapter Is Here Rather Than at the End

The published outline lists "Auctions, Portals, and Communities" second from last, after security and deployment. That is where it appears in the book whose contents page the outline was copied from, and it is the wrong place: these are not an afterthought to building a shop, they are *alternative shapes* a business can take, and the decision between them belongs in Part I alongside the other models.

So this chapter sits where it belongs. It is also the chapter that explains why the rest of the book assumes a fixed-price shop: the alternatives are genuinely harder, and it is worth knowing exactly how.

3.2 Letting the Buyer Set the Price

A fixed-price shop answers "what is this worth?" before the customer arrives. An auction refuses to answer and lets the buyers decide.

That is attractive when the seller genuinely does not know the price — a second-hand item, a one-off, something whose value depends on who wants it. It is unnecessary when the seller does know, which is why most commerce is not conducted by auction.

Four Auction Formats

English auction — ascending, open. The price rises until one bidder remains. What nearly everyone means by "auction".

Dutch auction — descending, open. The price starts high and falls until somebody accepts. Fast, and used for perishable goods where speed matters more than the last rupee.

Sealed-bid first-price — everyone submits one hidden bid; the highest wins and pays what they bid. Standard for tenders and contracts.

Vickrey auction (sealed-bid second-price) — the highest bid wins but pays the *second*-highest price. Its remarkable property is that bidding your true valuation is the best strategy whatever anyone else does, so there is nothing to be gained from strategising.

Why the second-price rule removes the guessing

Suppose an item is worth Rs. 10,000 to you.

In a *first-price* auction, bidding Rs. 10,000 guarantees you gain nothing even if you win — so you must shade your bid downward, and how far depends on guessing what others will do. Everyone does this, and everyone guesses.

In a *second-price* auction, bid Rs. 10,000. If the next highest bid is Rs. 8,000, you win and pay Rs. 8,000, gaining Rs. 2,000. If it is Rs. 11,000, you lose — and you are glad, because winning would have cost more than the item was worth to you. Raising your bid above your valuation can only make you win auctions you did not want; lowering it can only make you lose auctions you did want. Telling the truth is simply best.

Where you have met this already. The "proxy bidding" on consumer auction sites — you enter a maximum and the site bids on your behalf up to it — is a Vickrey auction wearing an English auction's clothes. The winner pays one increment above the second-highest maximum, not their own.

3.3 What an Auction Costs You in Engineering

A fixed-price product is a row that changes rarely. An auction is a small real-time system, and it brings four problems the rest of this book does not have to solve.

Problem	Why a fixed-price shop does not have it
Time is part of the state	A listing must close at an exact moment, unattended, and settle correctly whether or not anyone is looking. A product page has no deadline. This needs a scheduler, and a scheduler that did not run is a silent failure
Concurrent bids	Two bids arriving in the same instant must produce one consistent outcome. This is the classic lost-update problem and it needs a transaction and a lock, not a quick read-then-write
Sniping	A bid placed one second before closing wins without giving anyone a chance to respond. Fixes exist — extend the closing time on a late bid — and each changes the rules of the auction, which is a design decision rather than a bug fix
Shill bidding	The seller, or a friend, bids to push the price up with no intention of buying. Detecting it means analysing bidding patterns across accounts, and it is fraud rather than a feature gap

The lost update, concretely

Two bidders submit Rs. 5,000 at the same instant on an item currently at Rs. 4,500.

A naive implementation reads the current high bid, checks that the new bid beats it, and writes. Interleaved, both read Rs. 4,500, both pass the check, and both write Rs. 5,000 — the second overwriting the first. One bidder's money is now owed on an item they did not win, and the audit trail shows a single bid.

The fix is the one Chapter 10 uses for stock: do the check and the write in one atomic statement, inside a transaction, with the row locked. "Read, decide, write" is three steps, and anything can happen between them.

3.4 Holding Other People's Money

An auction site that merely introduces a buyer to a seller has introduced two strangers and left. Whoever moves first — sends the money, or sends the goods — carries all the risk.

Escrow is the operator's answer: it holds the buyer's money, tells the seller to ship, and releases the money when the buyer confirms receipt or a time limit expires.

Escrow and Dispute Resolution

Escrow is the holding of a payment by a trusted third party until the conditions of the transaction are met.

It makes the operator a custodian of money that is not theirs, which in most jurisdictions — Pakistan included — carries regulatory obligations (Chapter 22). It also makes the operator the **dispute resolution** authority, because somebody has to decide who is telling the truth when the buyer says the item never arrived and the seller says it did.

That adjudication cannot be automated away, and it is why marketplace commissions are higher than payment processing fees: the commission buys arbitration, not just plumbing.

Reputation as the cheap substitute, and what it cannot do

Escrow is expensive. The cheaper substitute is a **reputation system**: record how every past transaction went and let future counterparties read it.

It works, within limits, and the limits are worth knowing:

- **A new seller has no reputation**, so the system gives no protection in exactly the case where it is most needed.
- **Reputation can be bought** — a hundred small honest sales, then one large dishonest one. The score is an average; the fraud is a maximum.
- **Retaliation distorts it.** If each side rates the other, a buyer who reports a problem risks a bad rating in return, so both sides learn to stay quiet and the scores drift upward until they mean nothing.
- **It cannot be transferred.** Years of standing on one platform are worth nothing on another, which is exactly why the platform likes it.

3.5 Portals, Marketplaces and Communities

The three are often lumped together and should not be.

A **portal** aggregates *information* and sells attention — a price comparison site, a news aggregator with shopping links. It carries no stock, handles no payment, and earns from advertising or affiliate referral. Its engineering problem is ingesting other people's data and keeping it fresh, and its

commercial weakness is that it owns no transaction and can be cut out the moment a supplier decides to stop feeding it.

A **marketplace** aggregates *sellers* and hosts the transaction. It earns commission, and it is a shop in which the catalogue is written by strangers — which turns catalogue quality, duplicate detection and fraud into daily operational work rather than a one-time setup.

A **community** aggregates *people* who are there for each other rather than to transact. Commerce, if any, is secondary, and the usual mistake is assuming an engaged community will convert into a profitable marketplace. Many do not: the thing people valued was the absence of selling.

The two-sided launch problem

A marketplace is worthless to buyers with no sellers, and worthless to sellers with no buyers. On day one it has neither, and every new one faces this.

The strategies that actually work are all variations on *refusing to be two-sided at first*:

- **Be one of the sellers yourself** until there are enough others. You take on stock you did not want, and you compete with your own sellers, and it works.
- **Pick one narrow side and own it completely** — a single city, a single category — so that the market is small enough to fill.
- **Be useful to one side alone.** Offer sellers a tool worth having even with no buyers (stock management, a storefront), then introduce buyers once the sellers are already there.
- **Fake the thin side** by listing supply gathered from elsewhere. Common, legally risky, and the point at which several well-known companies acquired their reputations.

Notice that none of these is "build a better marketplace". The problem is not a product problem, and a student project that proposes a marketplace without addressing it has not yet met its hardest part.

Common Pitfalls

- **Implementing bidding with read-then-write.** Two simultaneous bids will eventually produce one lost bid, and it will be unreproducible when reported.
- **Closing auctions with a page request.** If closing happens when somebody loads the page, an auction nobody looks at never closes. It needs a scheduled job, and the job needs monitoring.
- **Treating escrow as a feature.** It is custody of other people's money and it is regulated. Chapter 22 is not optional reading if you intend to hold funds.
- **Believing a reputation score protects the first transaction.** It protects the hundredth.
- **Proposing a marketplace without a plan for the empty side.** This is the single most common fatal flaw in student project proposals for this course.
- **Assuming a community will monetise.** The engagement you measured may exist precisely because nothing was being sold.

Try It Yourself: Three shapes, and what each one must get right

No software yet. This is the last of the three analysis laboratories; Chapter 5 starts building.

Find one live example of each: an **auction**, a **marketplace** and a **portal**. They may be Pakistani or international, but you must be able to browse all three.

For each, record:

1. **Who takes the risk** if the goods never arrive — buyer, seller or operator? Find the actual policy page, do not guess. Quote it.

2. **Where the money is** between payment and delivery. Is there escrow? How long is it held, and what releases it?
3. **What the operator earns**, and from which side.
4. **One rule that exists because of a specific abuse.** Every mature platform has several — a bidding increment, a minimum feedback score, a holding period for new sellers, a limit on listing edits. Name one and work out what it is defending against.

Then, for the **auction** only: find the closing rules. Does a late bid extend the auction? By how much? Say which auction format that makes it, and whether it rewards bidding your true valuation.

Write up half a page ending with one sentence: of the three, which would be hardest to build, and which single problem makes it so?

What to notice. The rules that look arbitrary are nearly always scar tissue. Each one is a record of something that went wrong, and reading a mature platform's rules is the cheapest way to learn what will go wrong with yours.

Checkpoint — Test Yourself

1. In a Vickrey auction an item is worth Rs. 15,000 to you. Bids of Rs. 12,000 and Rs. 13,500 have been placed by others. What should you bid, what do you pay if you win, and why would bidding Rs. 18,000 be a mistake?
2. An auction site closes listings when a visitor loads the listing page. Describe precisely what happens to an auction for an unpopular item, and what the seller sees.
3. A marketplace has plenty of buyers and too few sellers. Give two strategies from this chapter that would address it, and name a cost of each.

Chapter Summary

- Auctions, portals, marketplaces and communities are alternative shapes a business can take, which is why this chapter sits in Part I rather than where the published outline puts it.
- The four formats are English, Dutch, sealed-bid first-price and Vickrey. Under the Vickrey rule, bidding your true valuation is optimal whatever others do — and consumer proxy bidding is Vickrey in disguise.
- An auction adds four problems a fixed-price shop does not have: time as state, concurrent bids, sniping and shill bidding. The first two are engineering; the second two are rule design.
- "Read the high bid, check it, write it" loses bids under concurrency. The check and the write must be one atomic operation.
- Escrow makes the operator a custodian of other people's money and the arbiter of their disputes. Reputation is the cheap substitute and fails exactly on the first transaction and the large one.
- Portals aggregate information, marketplaces aggregate sellers, communities aggregate people. Only the marketplace owns the transaction.
- Every two-sided market launches empty, and every strategy that works amounts to refusing to be two-sided at the start.

Review Questions

1. **(MCQ)** In a second-price sealed-bid auction, the optimal strategy is to bid: (a) slightly above your valuation; (b) exactly your valuation; (c) slightly below it; (d) it depends on the other bidders.

2. **(MCQ)** Which problem does a fixed-price shop share with an auction site? (a) Sniping; (b) shill bidding; (c) concurrent updates to one row; (d) scheduled closing.

3. **(Short)** Explain sniping, and give one rule change that discourages it together with the cost of that change.

4. **(Short)** Distinguish a portal from a marketplace, and say which is more easily cut out of its own business.

5. **(Short)** Give three distinct ways a reputation system can be gamed or rendered meaningless.

6. **(Applied)** Design the database handling for a single bid. Give the table columns you would need, the exact check that must happen atomically, and state what goes wrong if the check and the write are separate statements. You may write it as SQL or as precise prose.

7. **(Applied)** A student proposes a marketplace connecting Bahawalpur tailors with customers. Identify the two-sided launch problem in this specific case, choose one strategy from this chapter, and describe concretely what the first eight weeks would consist of.